Oracle Utilities Service and Measurement Data Foundation v25.10HOT

Product Fix Design

Bug 39187679

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Reported Issues

Description

One Day Closing Bill after the US stop date time fails with the application error:

```
2026-04-10 12:50:49.452-0700 [2709] ERROR
com.splwg.base.api.service.ServiceExecutionPolicy [userLogCorrelationId=af7879cc-8de0-40bc-b8e1-4068e44346ef] Encountered exception, session marked for rollback
java.lang.IllegalArgumentException: org.hibernate.QueryException: Incorrect query syntax [select ussp.id.servicePoint  from
com.splwg.d1.domain.usage.usageSubscription.entities.UsageSubscriptionServicePoint_Per ussp  where ussp.id.usageSubscription = :inUs   and ussp.id.startDateTime <= :referenceDateTime  and (ussp.stopDateTime is null or ussp.stopDateTime = :referenceDateTime)  /* Retrieve the list of service points linked to the usage subscription as of the reference date/time for the Closing Bill Condition */] at
org.hibernate.internal.ExceptionConverterImpl.convert(ExceptionConverterImpl.java:138)
```

Steps to Recreate

1. Create a device, device configuration, interval MC.
2. Create a service point and install the device on service point.
3. Create measurements on the interval MC from 03-30-2018 12 A.M to 07-25-2018 12:00AM.
4. Create usage calculation group with D2-Math Usage rule.
    a. **Add “Usage Calculation Rule Eligibility Criteria” using the Algorithm type D1-RCVEURCF (Retrieve Characteristic Value for Evaluation - Usage Rule EC Field).**
5. Create usage subscription with above service point and usage calculation group.

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6. Move the usage subscription to inactive status and set the end date/time as 07-24-2018 12 A.M.
7. Create a usage transaction for the following period -

   Start date/time 07-24-2018 12 A.M and End date/time 07-25-2018 12 A.M.
8. Observe that usage transaction throws an application error.

```
2026-04-10 12:50:49.452-0700 [2709] ERROR
com.splwg.base.api.service.ServiceExecutionPolicy [userLogCorrelationId=af7879cc-8de0-40bc-
b8e1-4068e44346ef] Encountered exception, session marked for rollback
java.lang.IllegalArgumentException: org.hibernate.QueryException: Incorrect query syntax [select
ussp.id.servicePoint  from
com.splwg.d1.domain.usage.usageSubscription.entities.UsageSubscriptionServicePoint_Per
ussp  where ussp.id.usageSubscription = :inUs   and ussp.id.startDateTime <=
:referenceDateTime  and (ussp.stopDateTime is null or ussp.stopDateTime =
:referenceDateTime)  /* Retrieve the list of service points linked to the usage subscription as of
the reference date/time for the Closing Bill Condition */]   at
org.hibernate.internal.ExceptionConverterImpl.convert(ExceptionConverterImpl.java:138)
```

Solution Design

In UsageSubscription_Impl.java, make changes to the method retrieveServicePointsAsOfDateTime(DateTime referenceDateTime) to convert 2nd HQL query to Prepared Statement SQL.

Test Case 1: Verify that usage is calculated for a single day closing usage transaction when the usage transaction starts on the usage subscription stop date/time and D2-Math Usage rule is used.

1. Create a device, device configuration, interval MC.
2. Create a service point and install the device on service point.
3. Create measurements on the interval MC from 03-30-2018 12 A.M to 07-25-2018 12:00AM.
4. Create usage calculation group with D2-Math Usage rule.
   a. **Add “Usage Calculation Rule Eligibility Criteria” using the Algorithm type D1-RCVEURCF (Retrieve Characteristic Value for Evaluation - Usage Rule EC Field).**
5. Create usage subscription with above service point and usage calculation group.

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6. Move the usage subscription to inactive status and set the end date/time as 07-24-2018 12 A.M.
7. Create a usage transaction for the following period -

   Start date/time 07-24-2018 12 A.M and End date/time 07-25-2018 12 A.M.

   Bill condition - closing
8. Verify that usage is calculated on the usage transaction.

Test Case 2: Verify that usage is calculated for a closing usage transaction that is created for two days.

1. Create a device, device configuration, interval MC.
2. Create a service point and install the device on service point.
3. Create measurements on the interval MC from 03-30-2018 12 A.M to 07-25-2018 12:00AM.
4. Create usage calculation group with D2-Math Usage rule.
   a. **Add “Usage Calculation Rule Eligibility Criteria” using the Algorithm type D1-RCVEURCF (Retrieve Characteristic Value for Evaluation - Usage Rule EC Field).**
5. Create usage subscription with above service point and usage calculation group.
6. Move the usage subscription to inactive status and set the end date/time as 07-24-2018 12 A.M.
7. Create a usage transaction for the following period -

   Start date/time 07-23-2018 12 A.M and End date/time 07-25-2018 12 A.M.

   Bill condition - closing
8. Verify that usage is calculated on the usage transaction.

Test Case 3: Verify that usage is calculated for an Interim usage transaction.

1. Create a device, device configuration, interval MC.
2. Create a service point and install the device on service point.
3. Create measurements on the interval MC from 03-30-2018 12 A.M to 07-25-2018 12:00AM.

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4. Create usage calculation group with D2-Math Usage rule.
    a. **Add “Usage Calculation Rule Eligibility Criteria” using the Algorithm type D1-RCVEURCF (Retrieve Characteristic Value for Evaluation - Usage Rule EC Field).**
5. Create usage subscription with above service point and usage calculation group.
6. Move the usage subscription to active status.
7. Create a usage transaction for the following period -

    Start date/time 07-23-2018 12 A.M and End date/time 07-24-2018 12 A.M.

    Bill condition - blank
8. Verify that usage is calculated on the usage transaction.

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